worldskills ASEAN Manila2025

Test Project

Session 3 - B

09 IT Software Solutions for Business

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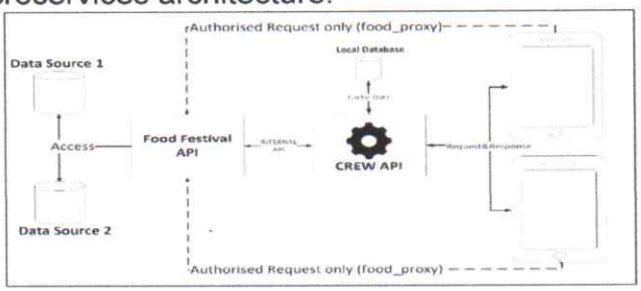
Introduction

Every year, a Food Festival is held in Manila, Philippines. As the host, the organizers are required to prepare a specially themed menu (Food of Day) for all visitors. Therefore, a large number of crew members are needed to support the entire program. Last year, a monolithic mobile application was developed to talk with the crew. However, the application faced performance issues due to the high volume of requests. As a result, you are required to follow software industry standards and develop a **new mobile applications** and **CrewAPI** which focusing on the Food Packaging Module using microservices architecture.



Infrastructure

FoodFestivalApi.exe – This is the provided API and host in isolated local environment, designed to address the performance issues of the old system and serve as the primary data source. mssql.sql and mysql.sql is the previous monolicthic database designs. [Note: you may cache the data in the database but FoodFestivalAPI remains primary data source]. food_proxy.exe –responsible for authenticating access to the FoodFestivalApi. assets folder – including some icons and other supporting resources.

FoodFestivalApi endpoint – Given API

[GET] api/server-log : Return access logs with formatting.
[GET] api/ingredients : Return response of ingredients list required by the FoodOfDay.
[GET] api/token : Set and Return the remember token to the client. This endpoints is important to authorize the request for the api/food, api/status, api/packaging.
[GET] api/food : Return response of FoodOfDay list.
[GET] api/status : Return the current status of food packaging activities.
[GET] api/packaging : Return steps of packaging based on the given packaging option.
* refer to **http://localhost:1234/api/hello** for parameters requirement and usage for each listed method.

CrewAPI (API to be developed)

- **Authentication in Crew API**

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Develop CrewAPI in Figure 1 which is middleware API between FoodFestivalAPI and mobile client. Retrieve and parse the login credentials (most are email) from the FoodFestivalApi [GET api/server-log]. For example: {{example@email.com}} (format in the api/server-log) should be trimmed to example@email.com. The credential might be blacklisted, blocked or invalid format — the credential cannot be used unless explicitly re-added by staff in the following logs. The purpose of this step is to identify which credential or email that is valid to login into the system. Store the credential into database (User table) for validation. Return result 'Authorized Credential' or 'Unauthorized Credential' when user login to the system. If the result is 'Authorized Credential', query [GET api/token] and set the remember token to the user with connection food_proxy.exe in the client mobile side.

- **[GET] api/ingredients/{FoodOfDayId}**

This endpoints in CrewAPI will retrieve the response from FoodFestivalApi (api/ingredients) and filter the responses based on food id. The overall structure and format of the response of this endpoint must remain consistent with the original api/ingredients response format from FoodFestivalApi

- **Deliverables**

Provide a file name, documentation.txt with following details: Base URL (Uniform resource locator) of the CrewAPI. All endpoints with responses developed in the CrewAPI. List all authorized Credentials stored in the User table.

Navigation Menu

A sidebar menu is accessible by tapping the top-left icon on the Food of the Day Page, Packaging Status Page or Info Page. The sidebar allows navigation between different pages. The remember token obtained from [GET api/token] should be displayed in the bottom of the sidebar. Tapping "Log Out" will return the user to the fresh Login Page.

Login Page

Using the dialog to display the authentication result of the login credential. Upon entering a valid login credential, user will be redirect to Food of Day Page. For testing purposes, the email admin@food.my is always allowed to log in. After user is authorized, get the remember token from the [GET] api/token.

Food of Day Page

This page displays the list of Food of Day from the FoodFestivalAPI [GET] api/food for the selected month and year. The color of calendar icon should be different based on the theme color of Food of Day. When user pulls down the list, request again from API and refresh the list. The list must be sorted ascendingly by date of the Food of Day

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and grey color for the past Food of Day (the date before today). When a food is selected, the user is redirected to the Food of the Day Detail Page. The "<" and ">" buttons allow users to navigate to the previous and next months, respectively.

Food of Day Detail Page

There is an info icon button. When it is pressed, a dialog appears showing food's details such as description and a list of main ingredients including their names and codes, ordered by position. ([GET] api/ingredients/{food_id}). All unique possible packaging options of the food are displayed in the list with their packaging time (3 decimal places) ascendingly (minutes). When a packaging option is clicked, a dialog appears showing the steps of selected packaging option. ([GET] api/packaging). The total count of unique packaging options should show in the end of the list. Example of packaging option code: (A X B), ((A X B) X C), (A X (B X C)), ...

Packaging Option

Each food ingredient is labelled with a Prepare time(minute), Scoop (number of serving produced after the prepare time), and the ingredients' stall position. Ingredient represents in this format : $\mathbf{A_1}$ (Ingredient A at position 1). The crew of the Food Festival uses 'Multi-Process Packaging' strategy to package ingredients by pairing any two adjacent ingredients until all ingredients are packaged into a food container. Adjacent Ingredients of **A** is {**B**} only; but Adjacent Ingredients of B is {**A**,**C**}; $\mathbf{A_1}$ and $\mathbf{B_2}$ can be packaged together which result as **($\mathbf{A_1}$ X $\mathbf{B_2}$)** or **Container N** ('X' indicate combination between two ingredient). Switching positions between ingredient is strictly prohibited. For example, $\mathbf{A_1}$ and $\mathbf{B_2}$ cannot be packaged as **($\mathbf{B_2}$ X $\mathbf{A_1}$).** Formula of Multi-Process Packaging in (A X B)

$$(A \text{ X } B)_{packaging_time} = (A_{time} * B_{scoop}) + (B_{time} * A_{scoop})$$
$$(A \text{ X } B)_{time} = A_{time}$$
$$(A \text{ X } B)_{scoop} = B_{scoop}$$
$$(A \text{ X } B)_{adjacent_ingredient}$$
$$= \{A_{adjacent_ingredient}\} + \{B_{adjacent_ingredient}\} - \{\text{packaged ingredients}\}$$
$$=\{B\} + \{A,C\} - \{A,B\}$$
$$=\{C\}$$

Total Packaging time of a packaging option = sum of the packing time from the steps of combination all ingredients in a container.

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Packaging Status Page

This page displays the current status of the packaging process fetched from the FoodFestivalApi ([GET] api/status) when user navigate to this page. Once all messages has been displayed, the list will stop updating and a "Refresh Status" button will appear/enable. Clicking this button will clear the current messages and reload the latest messages from the FoodFestivalApi. The status list will automatically scroll if the content exceeds the visible area. It will remain unchanged unless the user clicks the "Refresh Status" button or logout/exits from the application. If a message begins with "[Alert]", a dialog with the message will be displayed to the authenticated user immediately—even if they are on a different page.

Packaging Information Page

Display packaging_process.png and information of the diagram in this page. Provides explanation text to help crew to understand Multi-Process Packaging.

Mobile Application Deliverables

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